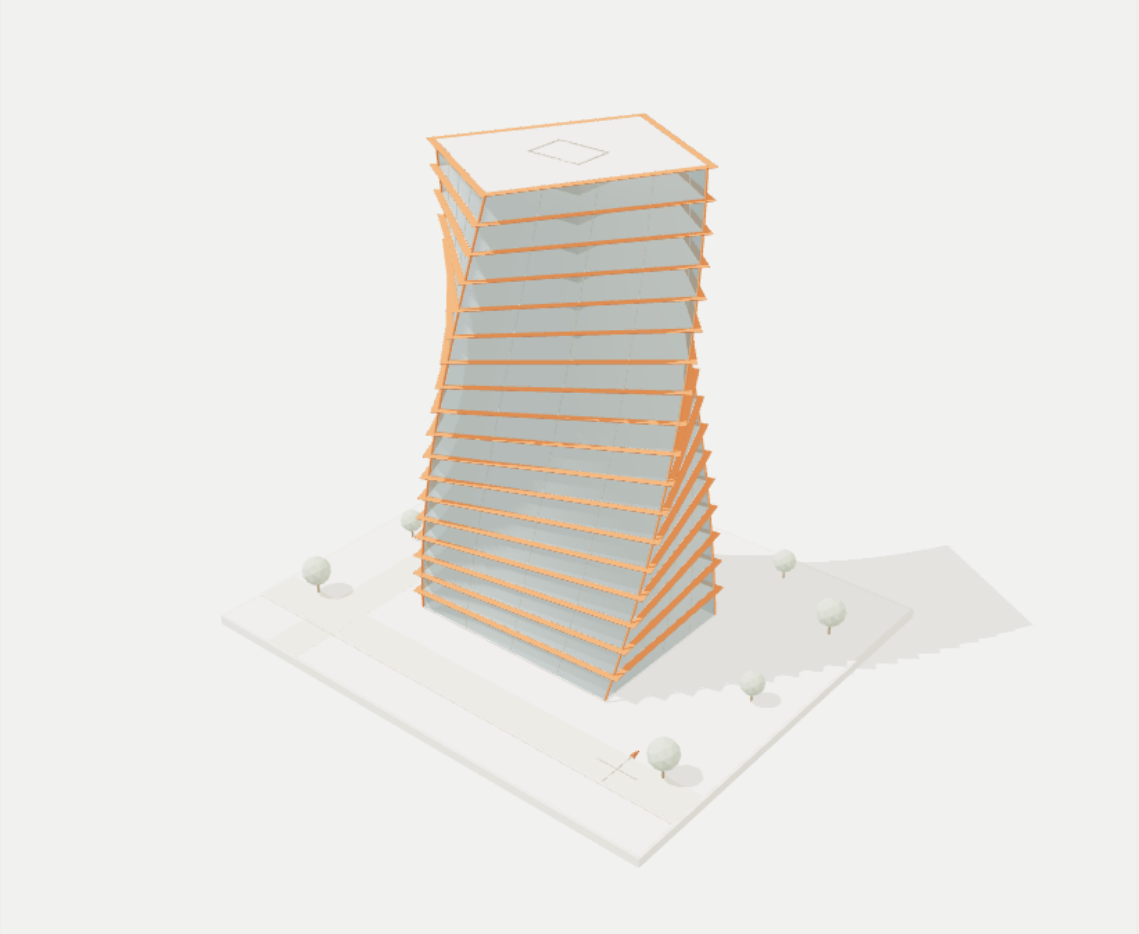
GENERATED FORM / CONSTRUCTION INFORMATION / SPATIAL DECISIONS

Aether

Form as a system.

A parametric building study connecting construction information, explicit design rules and alternatives that can be inspected.



Saatvik Rai

AI, software and computational design

LIVE WEB ENGINE

https://saatvik-rai-aether.vercel.app

18	60°	18%
STOREYS	TOTAL TWIST	TAPER

Concept geometry and authored service-zone checks.

The work behind Aether.

Construction-document systems, explicit requirements, and a model that makes their consequences visible.

RESEARCH QUESTION

How can explicit geometric rules and building requirements be used to generate, evaluate and revise spatial design alternatives?

MY TECHNICAL BACKGROUND AND CONTRIBUTION

My experience includes designing and implementing retrieval-augmented generation (RAG) pipelines, incorporating document processing, semantic chunking, embeddings and vector retrieval. I have also developed Python backend services and worked with interactive 3D building visualisation. These experiences motivated my interest in connecting information with spatial reasoning.

THE DESIGN PROBLEM

A construction requirement can lose its meaning when conditions, exceptions or cross-references are separated from it. Retrieving the relevant information alone does not establish which building elements it affects or how a proposed design change should be evaluated.

THE METHOD

In Aether, I explore this problem through synthetic geometry, explicitly defined requirements and computational checks. Each requirement is linked to its applicable geometric target, allowing alternative spatial arrangements to be evaluated and compared. The requirements and their mappings are manually defined; automatic regulatory interpretation is not claimed.

01 / SOURCE

Authored study brief / R02

Keep the service underside above the selected minimum.

02 / CONDITION

Minimum 2.40 m in occupied-zone

Plant-only bays require separate review.

03 / TARGET

service-route in occupied-zone

beam-01 determines the below-beam candidate geometry.

04 / CHECK

2.25 m < 2.40 m: R02 fails

Keep the clause, target and calculated result together.

Every view starts with the same vertices.

Floor count, scale and rotation define a repeatable family of forms.

01 / NORMALISE

$$t = i / n$$

Floor index i; floor count n.

02 / SCALE

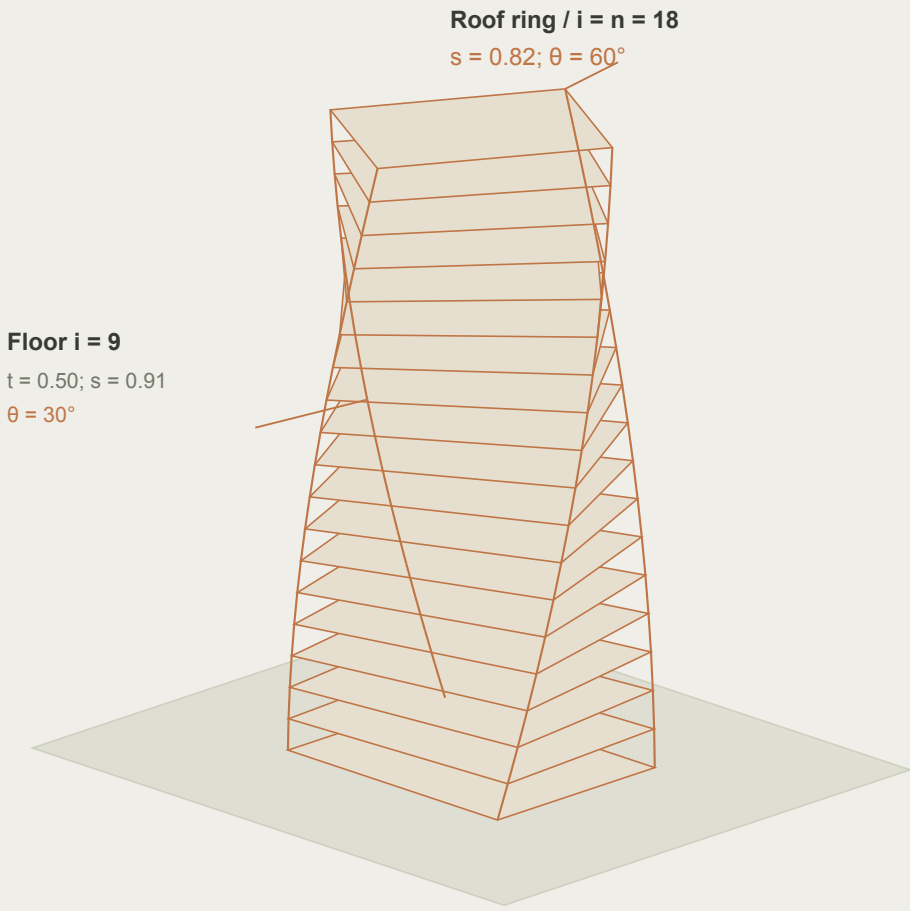
$$s = 1 - taper \times t$$

Each rectangle narrows with height.

03 / ROTATE

$$\theta = total\ twist \times t$$

Neighbouring rings connect through triangle faces.

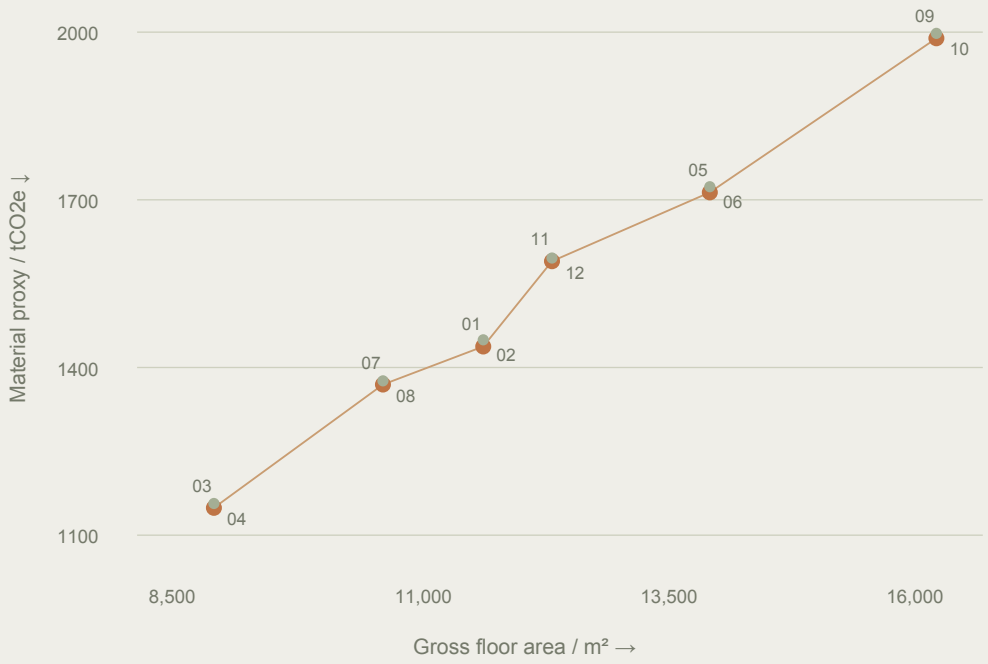


The engine supplies the viewport, measurements, sweep and OBJ export. Gross area includes core and circulation. Core fitting uses geometric clearance; structural and circulation adequacy are not established.

Baseline (18 storeys, 18% taper, 60° twist): 64.8 m high · 11,611 m² gross floor area · 6,749 m² envelope area.

The form is a choice.

A 3 × 2 × 2 sweep: 15, 18 or 21 storeys; 0% or 28% taper; 0° or 60° twist.



CASE	GFA / M²	PROXY / TCO2E	FRONT
01 15 / 0% / 0°	11,520	1,437.6	•
02 15 / 0% / 60°	11,520	1,449.5	-
03 15 / 28% / 0°	8,781	1,148.9	•
04 15 / 28% / 60°	8,781	1,156.7	-
05 18 / 0% / 0°	13,824	1,713.3	•
06 18 / 0% / 60°	13,824	1,723.4	-
07 18 / 28% / 0°	10,500	1,369.5	•
08 18 / 28% / 60°	10,500	1,376.1	-
09 21 / 0% / 0°	16,128	1,989.0	•
10 21 / 0% / 60°	16,128	1,997.8	-
11 21 / 28% / 0°	12,219	1,590.1	•
12 21 / 28% / 60°	12,219	1,595.9	-

PARETO FRONT

Non-dominated alternatives when maximising gross floor area and minimising the illustrative material-carbon proxy. No alternative on this front is better in both objectives simultaneously.

Under these two objectives, each untwisted case dominates its twisted counterpart. Twist preserves floor area while changing envelope area. This comparison does not decide spatial quality or architectural value.

Held fixed: base 32 × 24 m, floor height 3.6 m and shade depth 0.9 m. Material proxy factors are authored assumptions: concrete 320 kgCO2e/m³, facade 55 kgCO2e/m² and shades 18 kgCO2e/m².

Proxy scope: slabs, core walls, facade and shade allowances only. Excludes foundations, columns, reinforcement, MEP, interiors, construction and end of life. No solar or energy simulation is claimed.

A condition needs a geometric target.

The service zone is an authored test insert, used to make one information problem inspectable.

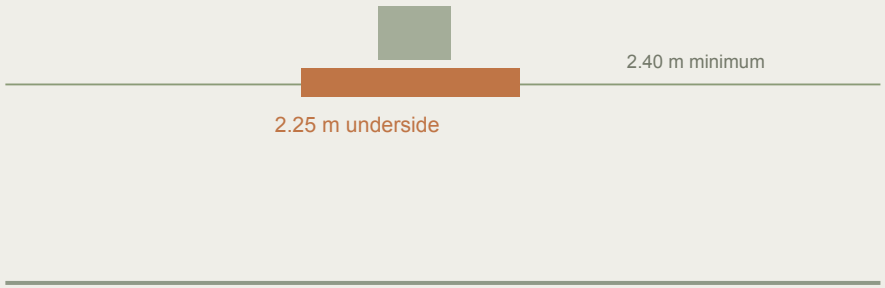
R02 / REQUIREMENT + APPLICABILITY

Service underside ≥ 2.40 m

Applies in occupied zones.

A fixed block can lose this condition. Keeping the authored paragraph together preserves it in six deliberately difficult boundary examples; a qualifier in another paragraph can still be lost.

SECTION / BELOW-BEAM CANDIDATE



R01 **Separation**

Keep the service clear of the beam, slab and zone bounds.

R02 **Headroom**

Check the lowest underside against the occupied-zone minimum.

R03 **Maintenance access**

Keep the declared access volume free while the requirement is active.

Authored geometry: 12 × 8 × 3.6 m zone; horizontal service envelope 0.60 × 0.35 m; 0.10 m separation. Vertical risers use a 0.60 × 0.60 m envelope.

Authored boundary examples, not a retrieval benchmark. Rectangular envelopes omit fittings, pressure loss and supports. Rules and mappings are explicit; no automatic regulatory interpretation, structural validation or statutory certification is claimed.

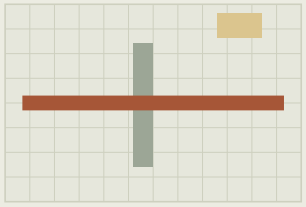
A smaller turn. A better candidate.

The derived bypass locates its centreline at 6.40 + 0.30 + 0.10 = 6.80 m.

EXPLORE THE LIVE ENGINE https://saatvik-rai-aether.vercel.app

01 / UNCHANGED

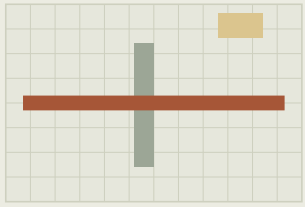
Direct reference



10.60 m Fails R01

02 / VERTICAL OFFSET

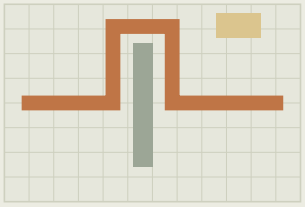
Pass below beam



11.75 m Fails R02

03 / PLAN DETOUR

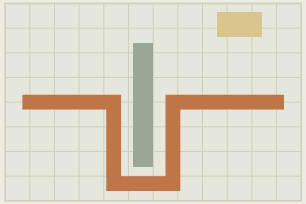
North local bypass



16.80 m Pass

04 / PLAN DETOUR

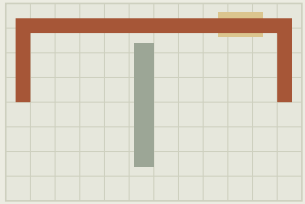
South local bypass



17.20 m Pass

05 / PERIMETER DETOUR

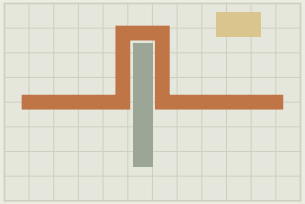
North perimeter



16.80 m Fails R03

06 / DERIVED FROM GEOMETRY

Clearance-derived bypass



16.20 m Pass

Baseline: the derived route passes at 16.20 m, 0.60 m shorter than the original north bypass. Three of six pass.

Model edit: with beam soffit 3.05 m, the below-beam route passes at 11.05 m with 2.60 m underside. Four of six pass.

Towards AI Agents That Retain Architectural Design Context

Can project memory recover the right conditions, decisions and checks across sessions and model revisions?

PROJECT MEMORY
Decisions + context

→

EVIDENCE AGENT
Clause + condition

→

GEOMETRY TOOLS
Inputs + check results

→

REVIEW + DESIGNER
Recheck + approval

A shared record would retain original clauses and exceptions, element IDs, model revisions, tool inputs/results and accepted decisions. Each agent would retrieve relevant records. Checkpoints would preserve the next task; summaries would point back to the records. Persistent agents are not implemented; the workflow and tests are proposed.

01 / RESUME

20 workflows / five sessions

Interrupt design tasks. Compare recent conversation, rolling summaries and versioned project memory under the same model and task budget.

MEASURE

Recovered decisions, missed constraints, outdated facts, unsupported answers and recovery time.

02 / RETAIN

30 reviewed clauses

Compare fixed chunks and linked records under one evidence budget. Include missing and conflicting conditions, exceptions and cross-references.

MEASURE

Lost conditions, wrong element links, unresolved references and appropriate requests for review.

03 / UPDATE

40 controlled edits

Compare names, element IDs and versioned dependencies against a full recheck and human annotations. Review disputed cases with designers.

MEASURE

Missed affected checks, stale passes and unnecessary rechecks.

I aim to investigate how persistent project memory, explicit architectural requirements and computational geometry can support the generation, revision and re-evaluation of spatial design alternatives. This research direction extends the questions explored in Aether and remains proposed work.